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Kelas : IF – 1

Rangkuman praktikum system terdesentralisasi dan terdistribusi

IMPLEMENTASI MICR SERVICE SEDERHANA DENGAN DOCKER DI AWS EC2

Aturan masuk

Info

ID aturan grup keamanan

Jenis

Info

Protokol

Info

Rentang port

Info

Sumber

Info

Deskripsi - opsional

Info

sgr-06f4552c279a1d049	TCP Kustom	TCP	8081	Kustom	Q	0.0.0.0/0	Hapus
sgr-0552b425d32c6c7b5	HTTP	TCP	80	Kustom	Q	0.0.0.0/0	Hapus
sgr-02f41767c424cbffb	TCP Kustom	TCP	15672	Kustom	Q	0.0.0.0/0	Hapus
sgr-06d508010f60a420a	SSH	TCP	22	Kustom	Q	0.0.0.0/0	Hapus
sgr-0774a170bb56c24d7	TCP Kustom	TCP	5672	Kustom	Q	0.0.0.0/0	Hapus
sgr-0d9475285b43ecb11	TCP Kustom	TCP	8082	Kustom	Q	0.0.0.0/0	Hapus
-	Semua ICMP - IPv4	ICMP	Semua	IPv4-Di ...	Q	0.0.0.0/0	Hapus
-	MySQL/Aurora	TCP	3306	IPv4-Di ...	Q 0.0.0.0/0	0.0.0.0/0	Hapus

Tambahkan aturan

Aturan masuk mengontrol apa jenis masuk yang diberikan untuk mengakses instance.

Aturan masuk

Info

ID aturan grup keamanan

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Sumber

Info

Deskripsi - opsional

Info

sgr-0558a02322333eb49	TCP Kustom	TCP	8081	Ku...	Q	0.0.0.0/0	Hapus
sgr-067b1987535c7076e	TCP Kustom	TCP	8082	Ku...	Q	0.0.0.0/0	Hapus
sgr-00718f302c1f2ace4	Semua ICMP - IPv4	ICMP	Semua	Ku...	Q	0.0.0.0/0	Hapus
sgr-0750a832e1d22f9df	SSH	TCP	22	Ku...	Q	0.0.0.0/0	Hapus
sgr-0afa6e308700ac706	UDP Kustom	UDP	123	Ku...	Q	0.0.0.0/0	Hapus

Tambahkan aturan

Batal

Pratinjau perubahan

Simpan aturan

Pembahasan : Protocol keamanan di setting

Untuk vm 1

SSH (TCP 22): Source 0.0.0.0/0

HTTP (TCP 80): Source 0.0.0.0/0

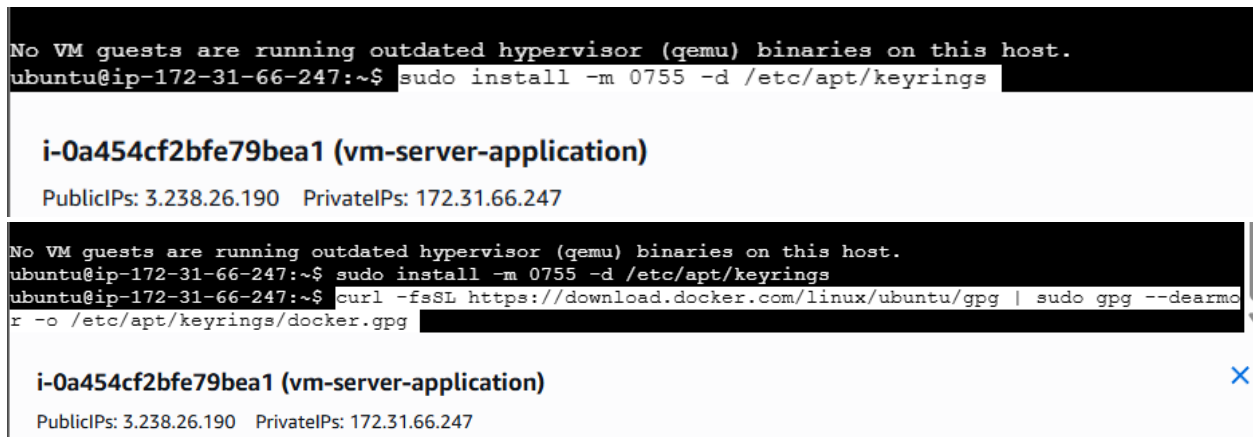
MYSQL (3306): Source 0.0.0.0/0

All ICMP -IPv4 (All): Source 0.0.0.0/0

Vm 2

SSH (TCP 22): Source 0.0.0.0/0

All ICMP -IPv4 (All): Source 0.0.0.0/0



```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-66-247:~$ sudo install -m 0755 -d /etc/apt/keyrings

i-0a454cf2bfe79bea1 (vm-server-application)
PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-66-247:~$ sudo install -m 0755 -d /etc/apt/keyrings
ubuntu@ip-172-31-66-247:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor
r -o /etc/apt/keyrings/docker.gpg

i-0a454cf2bfe79bea1 (vm-server-application)
PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247
```

Pembahasan : Instal docker dan docker compose dengan perintah

sudo apt-get update

sudo apt-get install -y ca-certificates curl gnupg

Setup GPG Key Docker (Jalankan per blok) Buat direktori keyring:

sudo install -m 0755 -d /etc/apt/keyrings

Download key (pastikan copy seluruh baris ini sekaligus):

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

Ubah permission file key:

sudo chmod a+r /etc/apt/keyrings/docker.gpg

```

ubuntu@ip-172-31-66-247:~$
ubuntu@ip-172-31-66-247:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
ubuntu@ip-172-31-66-247:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [41.3 kB]
Fetched 89.7 kB in 0s (222 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-66-247:~$

```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

Pembahasan : Tambahkan Repository Docker

```

echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo
"$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

```

```

Executing: /usr/lib/systemd/systemd-sysv-install enable docker
ubuntu@ip-172-31-66-247:~/microstore$ docker network create micro-net
109de902fd1ef56fd2798ea77ba398e44f394ad74a835020bebafe0fe7730fb
ubuntu@ip-172-31-66-247:~/microstore$ docker run -d \
  --name mysqldb \
  --network micro-net \
  -p 3306:3306 \
  -e MYSQL_ROOT_PASSWORD=password \
  -e MYSQL_DATABASE=microstore \
  -v $(pwd)/database/init.sql:/docker-entrypoint-initdb.d/init.sql \
  mysql:8.0
mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
ad9d782f3f87: Pull complete
8310b139f7c3: Pull complete
f34fb977c6ce: Pull complete
a2e746e7b2e3: Pull complete
750e74778c5d: Pull complete
75d963f8b5a9: Pull complete
acd41a114e50: Pull complete
b5f4bc928a89: Pull complete
082df7bb4f09: Pull complete
2583e052860a: Pull complete
3758af62b8ea: Pull complete
Digest: sha256:271aa4b4c84411d52790b340039dc6cea8637ab273546ff15639a77fae6b29f6
Status: Downloaded newer image for mysql:8.0
566ffae4a09b95b0337da3f59426174a8291df929903ef1263a8c7fd110f62c0
ubuntu@ip-172-31-66-247:~/microstore$

```

i-0a454cf2bfe79bea1 (vm-server-application)

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Pembahasan : git clone yang disediakan masuk cd microstore, membuat doocker network.
Kemudian docker run -d untuk menjalankan database

```
ubuntu@ip-172-31-66-247:~/microstore$ docker build -t image-product-service ./product-service
[+] Building 11.0s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 277B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> => sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 7.67kB / 7.67kB
=> => sha256:929b04d7c782f04f615cf785488fed452b6569f87c73ff666ad553a7554f0006 1.72kB / 1.72kB
=> => sha256:ee77c6cd7c1886ecc802ad6cedef3a8ec1ea27d1fb96162bf03dd3710839b8da 6.18kB / 6.18kB
=> => sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB
=> => sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4efea161b2b5a3d0e 40.01MB / 40.01MB
=> => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870
=> => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 1.26MB / 1.26MB
=> => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B
=> => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4efea161b2b5a3d0e 1.77kB / 1.77kB
=> => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 0.1s / 0.1s
=> => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 0.0s / 0.0s
=> [internal] load build context
=> => transferring context: 2.82kB
=> [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install --production
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:73a65613870ca4108516218aeb6172ed50c1802febce500145c5028c16e9b92c
=> => naming to docker.io/library/image-product-service
ubuntu@ip-172-31-66-247:~/microstore$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

```
ubuntu@ip-172-31-66-247:~/microstore$ docker run -d \
--name product-service \
--network micro-net \
-p 3001:3000 \
-e DB_HOST=mysqlldb \
-e DB_USER=root \
-e DB_PASS=password \
-e DB_NAME=microstore \
image-product-service
b939c6c4c85e0d9f87e7b28e26727de42ad64d043ebbb48d80784cb4a049f0ee0
ubuntu@ip-172-31-66-247:~/microstore$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

```
=> => writing image sha256:13a0618010ca1f08910210ac80122d30c10212b0c00f18c9320c10c9b320 0.0s
=> => naming to docker.io/library/image-product-service 0.0s
ubuntu@ip-172-31-66-247:~/microstore$ docker run -d \
--name product-service \
--network micro-net \
-p 3001:3000 \
-e DB_HOST=mysqlldb \
-e DB_USER=root \
-e DB_PASS=password \
-e DB_NAME=microstore \
image-product-service
b939c6c4c85e0d9f87e7b28e26727de42ad64d043ebb48d80784cb4a049f0ee0
ubuntu@ip-172-31-66-247:~/microstore$ docker build -t image-order-service ./order-service
[+] Building 8.6s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 277B                               0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  0.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 3.38kB                                  0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package*.json ./                                    0.1s
=> [4/5] RUN npm install --production                             6.9s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              1.1s
=> => exporting layers                                              1.1s
=> => writing image sha256:4b701d5b8fabdf6ec22230f2957d187ea9ea258238f9635ace21d070e38ecb0c 0.0s
=> => naming to docker.io/library/image-order-service             0.0s
ubuntu@ip-172-31-66-247:~/microstore$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

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```
ubuntu@ip-172-31-66-247:~/microstore$ docker build -t image-order-service ./order-service
[+] Building 8.6s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 277B                               0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  0.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 3.38kB                                  0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package*.json ./                                    0.1s
=> [4/5] RUN npm install --production                             6.9s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              1.1s
=> => exporting layers                                              1.1s
=> => writing image sha256:4b701d5b8fabdf6ec22230f2957d187ea9ea258238f9635ace21d070e38ecb0c 0.0s
=> => naming to docker.io/library/image-order-service             0.0s
ubuntu@ip-172-31-66-247:~/microstore$ docker run -d \
--name order-service \
--network micro-net \
-p 3002:3000 \
-e DB_HOST=mysqlldb \
-e DB_USER=root \
-e DB_PASS=password \
-e DB_NAME=microstore \
image-order-service
7b4b1e23284f15311df9c6ffe8dd1fc59350397a7973a615539f8b1f1dd27874
ubuntu@ip-172-31-66-247:~/microstore$
```

i-0a454cf2bfe79bea1 (vm-server-application)

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```
ubuntu@ip-172-31-66-247:~/microstore$ docker run -d \
--name api-gateway \
--network micro-net \
-p 80:80 \
-v $(pwd)/api-gateway/nginx.conf:/etc/nginx/nginx.conf:ro \
nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
1074353eac0d: Pull complete
25f453064fd3: Pull complete
567f84da6fbd: Pull complete
da7c973d8b92: Pull complete
33f95a0f3229: Pull complete
085c5e5aaa8e: Pull complete
0abf9e567266: Pull complete
de54cb821236: Pull complete
Digest: sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dd8db00020ad132295
Status: Downloaded newer image for nginx:alpine
697b5224565b8828499ce9c88daa89d24e634ccf593592b2f1ce4c028c872688
ubuntu@ip-172-31-66-247:~/microstore$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
697b5224565b   nginx:alpine  "/docker-entrypoint..." 18 seconds ago Up 17 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp
7b4b1e23284f   image-order-service  "docker-entrypoint.s..." About a minute ago Up About a minute 0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
b939c6c4c85e   image-product-service  "docker-entrypoint.s..." 4 minutes ago Up 4 minutes 0.0.0.0:3001->3000/tcp, [::]:3001->3000/tcp
566ffae4a09b   mysql:8.0    "docker-entrypoint.s..." 6 minutes ago Up 6 minutes 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
06->3306/tcp, 33060/tcp mysqlldb
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 3.238.26.190 PrivateIPs: 172.31.66.247

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Pembahasan : Build dan Run micro net, Product Service, Order Service, menjalankan API gateway Nginx dan tes jika muncul UP maka jalan dengan sempurna

```
Fetched 13.6 MB in 2s (5506 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
jq set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
ubuntu@ip-172-31-24-75:~$ curl http://3.238.26.190/products
{"success":true,"data":[{"id":1,"name":"Laptop Asus","price":"12500000.00","stock":10},{"id":2,"name":"Mouse Logitech","price":"350000.00","stock":50}]}
ubuntu@ip-172-31-24-75:~$ curl http://3.238.26.190/orders
{"success":true,"data":[{"id":1,"product_id":1,"quantity":2,"total_price":"25000000.00","product_name":"Laptop Asus"}, {"id":2,"product_id":2,"quantity":5,"total_price":"1750000.00","product_name":"Mouse Logitech"}]}
ubuntu@ip-172-31-24-75:~$
```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

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Pembahasan : Pengujian dari VM 2 client dengan cara curl http://3.238.26.190 /products , http://3.238.26.190 /orders selesai

Kesimpulan : Praktikum ini mendemonstrasikan implementasi arsitektur microservices menggunakan Docker di AWS EC2 dengan memisahkan fungsi bisnis ke dalam container yang terisolasi. Melalui penggunaan API Gateway (Nginx) sebagai pintu masuk tunggal dan Docker Network sebagai media komunikasi antar-layanan, sistem ini mampu mengarahkan trafik secara efisien dari sisi client ke layanan produk maupun pemesanan.

